

Machine Learning Models for Horse Racing Betting: A Literature Review

Introduction

Horse racing betting has attracted significant attention from data scientists and gamblers alike, especially with the rise of betting exchanges like Betfair that provide rich real-time data. Researchers and practitioners have explored whether machine learning (ML) can find an **edge** in these markets by predicting race outcomes or price movements more accurately than the crowd. Early successes such as Bill Benter's algorithmic approach in Hong Kong (which reportedly earned him hundreds of millions) showed the potential of data-driven betting ¹. However, modern betting exchanges present a highly competitive and arguably efficient market, so any profitable strategy must intelligently leverage data and withstand practical constraints. This review surveys the literature on ML models applied to horse racing betting, with an emphasis on **Betfair-style data** (exchange odds, traded volumes, market depth, Betfair Starting Price, etc.), **profitability outcomes**, and **practical implementation considerations**. Key topics include the types of features used from exchange data, common ML modeling approaches, evidence of profitability or market efficiency, ensemble and rule-based strategies for usability, pitfalls (e.g. data leakage, execution friction), and how models are adapted for real-world betting (from personal trading bots to group "punters club" decision support). The goal is to summarize findings from academic papers, preprints, and applied studies that have *tested performance against market benchmarks or profitability*, providing guidance on what has (and hasn't) worked in this challenging domain.

Betfair Data Features Used in Horse Racing Models

A first step in horse racing ML is understanding what data features are available and predictive. **Betfair exchange data** offers a granular view of the betting market that goes beyond final odds or simple race results. Studies have leveraged a variety of feature types from Betfair and related sources:

- **Odds and Price Dynamics:** The most fundamental features are the odds for each horse, recorded over time up to race start. This includes the **Betfair Starting Price (BSP)** (the exchange's closing odds) and snapshots of odds at various intervals before the off ². Many models use the entire **odds time series** or derived metrics (e.g. the amount of late odds **drift or steam** for a horse). Sudden odds movements (plunging "steamers" or lengthening "drifters") are of particular interest as potential signals of new information. For example, one analysis noted instances of horses trading at long odds (e.g. 10.5) shortly before the race and cratering to a BSP of 2.8 – an extreme "steam" indicative of heavy insider-backed support ³. Features capturing such **last-minute volatility** or **odds velocity** can be predictive if they reflect informed betting.
- **Traded Volume and Money Flow:** Betfair provides data on **matched volumes** at each price. Models often include features like the total amount matched on a runner (overall or in specific time windows) and **weight of money** (the volume waiting on the back vs lay side of the order book). Such features gauge market sentiment and liquidity. For instance, Dzalbs et al. (2017) built a trading

strategy predictor using Betfair API data with **price and volume** information as inputs ⁴. High traded volume or rapid increases in volume can signify confidence behind a price move, while thin volume might indicate an illiquid market prone to abrupt shifts.

- **Market Depth and Order Book Features:** Beyond matched volume, advanced approaches use the full **limit order book** state – i.e. the available back and lay offers at different odds. Gonçalves et al. (2025) argue that **market depth data** is critical for short-term price forecasting on Betfair ⁵. Their models ingest the multi-level order book (prices and sizes) to predict immediate price movements. Features like **best back/lay prices**, **market spread** (difference between best lay and best back odds), and how these change can serve as indicators of market efficiency or impending moves ⁶. Rapidly tightening spreads and rising volumes at a given price, for example, might precede an odds change.
- **Runner and Race Metadata:** While exchange markets provide pure betting data, successful models often **combine it with traditional racing data** (the kind a handicapper uses). This includes horse performance metrics (recent finishes, speed ratings), horse/jockey/trainer statistics, track conditions, etc. Dzalbs et al. incorporated **fundamental data** like horse form and past running history in addition to technical market data ⁴. Similarly, ensemble models built on Australian and UK racing data included extensive race information from form guides (age, weight, going, trainer form, etc.) alongside any available odds data ⁷ ⁸. These features provide contextual information that pure market signals might miss, potentially helping an ML model identify mispriced horses (e.g. a horse with strong track conditions advantage that the market hasn't fully priced in).
- **Aggregate Market Indicators:** Some research derives features that describe the **overall market dynamics** for a race. For example, **market "tightness"** or overround (how close to 100% the sum of implied probabilities is) can indicate how efficient or competitive the pricing is ⁹. **Inter-runner relationships** like the odds rank of a horse (favorite, second favorite, etc.) or the gap between a horse's odds and the top rival are also used. These can capture effects like favorite-longshot bias or identify which races are more upset-prone. In one tutorial, selections were categorized by BSP ranges (odds-on, moderate, longshot, etc.) and analyzed for different market behaviors ¹⁰ ⁹, recognizing that a 2.0 odds horse behaves differently in the market than a 50.0 odds outsider.

In summary, Betfair-based models draw on a rich feature set: **temporal odds patterns, traded volume curves, order book snapshots, price-derived signals (volatility, momentum), and traditional form guide data**. The combination of **real-time market features with static horse attributes** is a common theme in profitable models, as it allows capturing both *what the market thinks* and *what perhaps it might be overlooking*.

Machine Learning Approaches Applied to Horse Racing

Researchers have applied a wide spectrum of machine learning techniques to horse racing betting, ranging from straightforward classification models to sophisticated time-series deep learning. Broadly, we can categorize approaches into a few groups:

- **Outcome Prediction (Classification/Regression):** Many studies treat horse racing prediction as a classification problem (win vs lose for each horse) or a regression (estimating win probability or expected finish position). Traditional algorithms like logistic regression and linear discriminant

analysis (LDA) were used in early works, but modern studies often use more powerful classifiers. For example, Lessmann et al. (2010) applied a **Random Forest classifier** to Hong Kong race data and found it outperformed older methods (like a logistic model) in both accuracy and betting returns ¹¹. Similarly, Borowski et al. (2021) tested six algorithms (CART, Glmnet, XGBoost, Random Forest, Neural Network, and LDA) on Polish racing; interestingly, the best predictive performance came from **LDA and Neural Network models**, which correctly identified around 41% of winners in their test (significantly better than random ~8% in a 12-horse race) ¹². Ensemble tree methods like **XGBoost and LightGBM** have become popular for structured racing data, often winning tabular-data competitions and academic evaluations due to their ability to handle many features and interactions. For instance, an upcoming 2024 study by Khanh et al. built a **stacked ensemble** of six algorithms (including LightGBM, XGBoost, CatBoost, etc.) to forecast race winners, leveraging large datasets from Australia and the UK ⁸.

- **Ranking and Probability Estimation:** Instead of predicting a binary outcome for each horse independently, some approaches model the *entire field* outcome. This can be done via **ranking algorithms** or by outputting a probability for each horse and using that as a ranking. The Random Forest model by Lessmann was adapted to account for the competitive nature of the race (effectively producing a ranking of horses' chances) and evaluated with metrics like Normalized Discounted Cumulative Gain (NDCG) and betting profitability ¹¹. Another approach is using **multinomial models or pairwise comparisons** to derive odds. For example, one could use an Elo or Bradley-Terry model to rate horses, or a Plackett-Luce model to output a probability distribution over placements. These are less common in recent ML literature compared to classification/ensembles, but they offer a more **probabilistic framework** aligned with odds-making. Some studies also predicted **margin of victory or running time** as a regression, then inferred the order – for instance, a data scientist on Medium detailed a model predicting each horse's finish time as a way to rank horses in a race ¹³. Such regression-to-rank approaches can incorporate rich information (predict time given distance, conditions, etc.) and naturally handle races with many runners.
- **Time-Series and In-Play Models:** A distinct line of research looks at **predicting odds movements and trading opportunities** on exchanges, treating it akin to a financial time-series problem. Here, the target might be *short-term price direction* (will a horse's odds shorten or drift in the next minute?), allowing a trading bot to place a bet and then hedge or trade out for profit. Advanced ML techniques have been employed: Gonçalves et al. (2019, 2025) used **Deep Neural Networks, LSTMs, and CNNs** with novel convolutional attention mechanisms to forecast very short-term price changes in Betfair horse markets ¹⁴ ⁵. Their 2025 work introduced a specialized **bi-dimensional convolutional recurrent network** to process Betfair order book snapshots over time, enabling a model to spot patterns like momentum or mean reversion in odds. Another study by Dzalbs et al. (2017) combined a **feed-forward neural network** (for pattern recognition) with **Cartesian Genetic Programming** (for evolving trading rules) to predict pre-race price movements ¹⁵. These time-series models often frame the task as a classification (e.g. "price will go down, stay, or go up") and use sliding window features of odds and volumes. **Reinforcement learning** could also be applied in this setting, though most literature so far uses supervised learning on historical data rather than training an agent via simulated betting.
- **Ensembles and Hybrid Models:** Ensemble learning is prominent in recent horse racing research, both for improving prediction accuracy and for combining different information sources. As mentioned, stacking and boosting methods have shown strong performance ⁸. Ensembles can

also blend fundamentally different approaches: e.g. a model that analyzes **race form data** can be ensembled with another that analyzes **live market data** to cover both angles (essentially a hybrid of handicapping and market-watching). Some studies explicitly use **committee or voting approaches** to decide bets – for example, if multiple models agree a horse is undervalued, that selection is stronger. The idea of a second model or rule acting as a **filter** (sometimes described in forums as a “gate” model) is also applied in practice to increase precision. While formal literature on gated ensembles in racing is sparse, the concept is to train one model to generate candidate bets and another to predict which of those are actually worth betting (perhaps to control false positives). This falls under the broader strategy optimization discussed later.

- **Other AI Techniques:** A few niche approaches have been explored. **Fuzzy logic systems** have been used to handle the uncertainty in racing data – for instance, a 2015 study built a fuzzy inference model using inputs like horse speed and past performance, achieving about 75% accuracy in their dataset ¹⁶. This interpretable rule-based approach is akin to expert systems and can be more transparent for users. There have also been **genetic algorithms** for feature selection or weight optimization, and agent-based simulations (like the BBE model from Bristol) to train algorithms in a synthetic environment ¹⁷. The BBE work by Cliff and colleagues essentially created a simulated betting exchange with simple bettor agents, then used ML (XGBoost) to discover profitable betting policies within that sim ¹⁷. This is a unique crossover of AI, providing a controlled sandbox to develop strategies that might later be applied to real markets.

In summary, **classification and ensemble tree-based models** are the workhorses for predicting race outcomes, often augmented now by neural network approaches as data availability grows. Meanwhile, **time-series deep learning** is pushing the frontier in modeling the *market itself* (prices and volumes over time) to find trading edges. It's common to see researchers try multiple algorithms and report that ensemble or nonlinear methods (Random Forest, XGBoost, Neural Nets) outperform linear baselines in terms of both accuracy and betting returns ¹¹ ¹⁸. However, raw predictive power doesn't always translate to profit – which we examine next.

Profitability vs. Market Efficiency

A central question in this domain is whether these ML models can actually beat the betting market – i.e. achieve positive returns on investment (ROI) after accounting for odds and commissions. The literature shows mixed results, reflecting the **high efficiency** of mature betting markets like Betfair. Key findings include:

- **Markets Are Hard to Beat:** Several studies underscore that horse racing markets, especially on exchanges, are very efficient in incorporating information. Tondapu (2024) analyzed Betfair's UK horse racing odds and found “a remarkably high level of informational efficiency”, with price changes showing light tails, rapid decay in autocorrelation, and quick mean reversion ¹⁹. In plain terms, prices move quickly to reflect any new information, and there's little long-term momentum to exploit. This aligns with the idea that **Betfair Starting Price (BSP)** is an accurate proxy of the true probability – so beating BSP consistently is difficult. Indeed, experienced bettors often use beating the BSP (getting better odds than BSP on average) or beating the **closing line value (CLV)** as a litmus test for an edge ²⁰. Analyses confirm that “a disciplined approach can yield positive returns when consistently beating the Starting Price (SP)” ²⁰. Conversely, if a model's picks do not beat the closing odds on average, any short-term profit is likely due to luck.

- **Positive ROI Demonstrations:** Despite efficiency, there are documented cases of ML models yielding profitable results, at least historically:
- **Dzalbs et al. (2017)** reported an impressive **+83% ROI** over a one-month backtest of their automated trading agent on Betfair ²¹. This agent used neural networks and genetic programming to forecast price moves and executed a strategy (backing and laying) with a fixed liability stake. The ROI was for a short period (one month of UK races in 2016) and was *simulated*, but it showed the potential of a data-driven trading system. It's worth noting such high ROI is likely on small volume and could regress over a longer sample, but it indicates that **inefficiencies do exist** briefly (perhaps in lower-tier races or split-second in-play situations).
- **Gonçalves et al. (2019)** (as summarized by KTH [41]) developed models (DNN, LSTM, CNN) to predict pre-race odds movement and tested a fully automated trading algorithm for 30 days on Betfair. They reported **profitable results for all models**, with the CNN-based strategy earning the highest profit over that month ¹⁴. This suggests that their short-term predictions had enough edge to overcome transaction costs consistently, at least in that trial period.
- **Lessmann et al. (2010)**, focusing on the Hong Kong tote (a very competitive market), showed that a Random Forest model could outperform the market odds when paired with an optimal betting strategy. Using a **Kelly criterion** staking, their model achieved "*significant profits*" in testing ²². This is notable because Hong Kong betting has a high takeout (~17%), so showing profit means the model identified value where the public odds were notably off. Their success highlights the importance of both a good model and a sound wagering strategy (Kelly bet sizing maximizes growth from an edge).
- **Borowski et al. (2021)** found that in Polish flat racing, certain models could achieve **positive ROI**. Their best model (LDA on a combined dataset) yielded the highest ROI for Win bets, indicating that "*profitable betting could be achieved under certain conditions*" ²³. The correct bet rate was around 41%, and while ROI figures weren't explicitly quoted in that summary, the implication is that with proper selection of races (perhaps focusing on specific race types or odds ranges), an edge existed in that market. This might reflect that less-liquid or smaller markets (like Polish racing) have more pricing errors for a model to exploit, compared to major markets.
- Outside academia, anecdotal reports on forums and blogs also claim positive results. For instance, a Reddit user who backtested a UK horse racing model over 9 years noted that the model's selections "*consistently outperformed the probabilities implied by BSP*", yielding a steady positive ROI after Betfair commission (implying genuine predictive edge) ²⁴. While not formally published, such accounts align with the idea that **value betting** – picking horses whose true win probability is higher than the odds imply – can work if done rigorously over a large sample.
- **Marginal or No-Profit Findings:** On the flip side, many experiments show how difficult it is to maintain an edge once realistic constraints are applied:
- Santos (2019) built a trading agent that combined simple arbitrage logic with ML predictions in Betfair win markets. It initially showed promise (improved performance metrics) but ultimately **still made a loss in the long run** ²⁵ ²⁶. The culprit was practical execution – the strategy relied on capturing moments of mispricing (back high, lay low for a same horse), but often *the necessary odds were not actually available in sufficient volume*. In other words, **liquidity issues** and the **inability to get matched** at the desired price prevented profit, even though on paper the opportunities existed. This highlights a common pitfall: a theoretical edge might vanish in real trading due to market frictions.

- Øvregård (2008) applied neural networks to in-play tennis betting (another exchange market) and even introduced a custom cost function to optimize for betting outcomes. The result was a very modest **+0.21% yield** ²⁷. While positive, such a razor-thin margin would likely be eaten by commission – essentially break-even. It underscores that ML models, even when tuned, often end up only slightly better than coin-flip in efficient markets.
- Gupta and Singh (2024) achieved **93% accuracy** in predicting race winners on an Indian dataset using Random Forest ²⁸, but high accuracy doesn't guarantee betting profit if the model is largely reiterating what the odds already tell us. A model that picks the favorite every time might get ~30-40% winners right (accuracy) in horse racing, but it will almost certainly lose money because favorites are overbet. In Gupta's case, without details on odds or ROI, it's unclear if their high accuracy translated to an edge or was partly a result of data quirks. This exemplifies that **predictive accuracy and betting efficiency are not the same** – a model must find *value*, not just winners, to be profitable.
- **Market Efficiency and Closing Line Value:** A recurrent theme is that the **closing odds (BSP)** represent the collective wisdom of the market and are tough to beat. Betfair's exchange odds have no built-in bookmaker margin, so BSP is an even sharper benchmark than bookmaker SP. Studies have observed that any readily available information (public data) tends to be reflected in the odds by race time. For example, if a horse drifts from 5.0 to 10.0, there was likely a real issue (negative info) causing it; betting it just because the price is higher is not a value strategy because the horse's true chances probably dropped. One tutorial explicitly demonstrated this: after identifying the top 10 biggest plunges in odds in a sample of races (horses with huge late steam), they checked what would happen if you bet those horses at BSP. Surprisingly, that specific subset showed a ~+70% profit on turnover ²⁹ – but this was a tiny sample and more of a curiosity (some heavily-backed horses still won at decent odds). When expanded to a broader sample of "large odds plunges" cases, the result was essentially no profit: *"there's not much of a pattern here...the value has been sucked dry from the market on these selections."* ³⁰. In other words, once a horse steams in price, the final odds typically adjust to a point where neither backing nor laying such movers yields an edge on average. **The market knows** – by the time you react, it's usually too late.

In aggregate, the literature suggests **sustainable profitability is possible but requires genuine insight beyond what the market already prices in**. Small edges (a few percent ROI) are more realistic than large windfalls. Many academic studies that claim high profitability often either test on short time spans or ignore certain costs, so their results should be taken cautiously. The more convincing demonstrations are those that measure performance *against BSP or closing odds*. For instance, models whose implied probabilities show a better Brier score or log loss than the implied probabilities of BSP are truly adding information. Some works implicitly do this (Lessmann's outperforming market odds, Borowski's ROI > 0, etc.) and those are the promising ones. It's also notable that **in smaller or less efficient markets**, ML models have more room to shine – e.g., Borowski in Poland, Pudaruth in Mauritius (who achieved 58% win prediction accuracy, beating expert tipsters' 44% ³¹), or other studies on secondary markets. In major markets like the UK, the edge might be slimmer, but even a 2-5% edge can be very lucrative at scale and with low variance if real.

Ensemble Strategies and Explainability for Group Use

To improve reliability and trust in model outputs – especially important if the system is used by a group of human bettors (a punters club) – researchers have looked at **ensemble methods, hybrid strategies, and adding interpretability**:

- **Stacked and Ensemble Models:** Using multiple models in concert can yield more stable predictions. The 2024 study by Nguyen et al. is a prime example: they built a **stacked ensemble** of six ML algorithms to predict winners, combining tree-based models and a neural network to leverage different algorithm strengths ⁸. Ensemble predictions can reduce the variance and error of single models. Moreover, ensembles allow **diversity in features or data sources** – e.g., one model might focus on live Betfair data while another focuses on historical performance factors, and the ensemble can blend these. In practice, ensembles have empirically improved accuracy, but they also lend themselves to more **robust betting strategies**. If three different models all flag the same horse as undervalued, one can bet with more confidence (some punters clubs explicitly take consensus of multiple methods/tipsters to decide bets).
- **Strategy Optimization and Filters:** Beyond predicting outcomes, some research optimizes *which bets to actually place* to maximize profit or ROI. This often involves imposing rules or secondary models on top of raw predictions. For instance, Khanh et al. (2024) experimented with different betting strategy objectives: one approach maximized prediction accuracy, while another explicitly aimed to **maximize ROI by selecting longer-odds underdogs** ³² ³³. They found the **“Optimize ROI for Underdogs” strategy was most profitable despite lower accuracy**, suggesting that focusing on value (often in higher odds horses) beats chasing a high win percentage ³⁴ ³⁵. This illustrates a rule that could be used in a punters club context: instead of simply taking the model's top probability horse (which will often be the favorite and not a good bet), one might filter or weight selections towards those with the largest implied value edge (which frequently are outsiders). Similarly, practitioners often apply a **minimum value threshold** – e.g. only bet if the model's estimated win probability exceeds the implied probability of the available odds by, say, 5% or more. This kind of rule-based filter (no matter what the model says, skip low-margin bets) can improve betting ROI and also aligns with human intuition to avoid “maybe value” bets.
- **Secondary Models (Gating) and Ensembles of Experts:** In more advanced setups, one model might generate an initial list of potential bets and another model (or set of rules) acts as a gatekeeper. For example, a forum post described a two-stage system: *“the model basically scores pre-race data and makes selections, a second model acts as a gate and decides the picks”*. While informal, this approach resonates with the idea of **ensemble of experts** – one expert proposes bets, another vetos or confirms them. We don't see this explicitly in academic papers, but a similar concept is cascading models: first predict probabilities, then have a model that predicts if a given probability/odds gap is worth betting. The **explainability** benefit here is that the second model can be simpler or based on clear criteria (like “bet if edge > X and conditions Y,Z hold”), making the final decision more transparent to users.
- **Human-Interpretable Models and Factors:** In a punters club or social betting setting, participants may want to understand *why* a model likes a certain horse. Pure black-box models can be met with skepticism (“Why is the algorithm picking Horse #7? That horse looks out of form!”). To address this,

some researchers either stick to more interpretable methods or they provide explanations for complex models:

- One approach is using **weighted factor models** (which are inherently explainable). Pudaruth et al. (2013) took this route by assigning weights to various factors (jockey, horse form, draw, odds, etc.) and combining them into a probability prediction ³⁶. While this was not a machine-learned weighting (they likely tuned it manually or via simple optimization), it produced good results and is easy to explain ("Horse A is top because it has the highest weighted score considering X, Y, Z"). In modern ML, one could use techniques like **SHAP (Shapley values)** or feature importance from tree models to explain each prediction – e.g., telling punters *"Horse #7 is selected because our model noted its recent speed figure and dropping weight, combined with a significant late market backing (odds fell from 15.0 to 9.0), offer value at the current price."* Such explanations blend data-driven insight with racing logic.
- Another strategy for explainability is to integrate **rule-based filters** that the group agrees on. For example, a punters club might decide they will not bet on races with very heavy favorites, or they will avoid certain race types the model historically struggles with (maybe maiden races or very large fields). These rules can be informed by model analysis but framed in a way the group finds reasonable. By incorporating these constraints, the model's output becomes more aligned with human bettors' comfort, even if it sacrifices some theoretical profit.
- **Automated Betting Systems vs. Decision Support:** The literature also distinguishes between fully automated systems and those that provide guidance to human bettors. In fully automated **trading bots**, explainability is less of an issue than raw performance – the model can trade hundreds of small edges without human intervention. Many of the high-ROI studies (Dzalbs, Gonçalves) fall in this category, deploying algorithms directly on Betfair via API ³⁷. For **group betting (punters clubs)**, the model usually plays an advisory role (e.g., recommending bets or presenting a ranked list of horses), and the humans make final decisions and stake adjustments. In such cases, presenting the output in an understandable way is crucial. A common solution is to provide a **confidence rating or "bet strength"** for each tip, which could be derived from the model's estimated edge or ensemble agreement. This helps club members feel in control – they might choose the highest confidence bets or discuss lower confidence ones rather than blindly following an algorithm.

In essence, combining models (and even combining model insight with human judgment) appears to be a robust recipe. Ensembles tend to perform better and also offer a way to incorporate multiple perspectives (speed figures, form analysis, market signals, etc.). When it comes to **maximizing profit**, research suggests favoring strategies that differ from the crowd (e.g. value in under-bet longshots) ³² ³³, even if it means a lower strike rate – something human bettors may need to emotionally prepare for. This is a key insight for punters clubs: a model that picks many losers at high odds can still be profitable long-term, but the group must understand and embrace the strategy. Tools like thorough backtesting, and clearly communicating the **expected distribution of wins and losses**, become part of the process to build trust in the model.

Pitfalls and Failure Modes in the Literature

Machine learning efforts in horse racing are prone to various pitfalls, some common to predictive modeling and some specific to betting. Recognizing these failure modes is important to avoid overstating results or deploying flawed systems:

- **Data Leakage:** Perhaps the most dangerous pitfall is using information in training that wouldn't be available before race time – inadvertently giving your model hindsight. In horse racing, an easy way to leak information is through odds. For example, including the BSP or near-off odds as features when predicting winners will make a model seem uncannily accurate, but in reality those odds *contain* the collective prediction of the outcome. Any model that has access to the final market odds will simply learn the market's probabilities (hence very high accuracy, as seen in some papers boasting 90%+ accuracy ²⁸). That's trivial and useless for betting since you can't know those odds in advance. Even more subtly, including **mid-race information** (like in-play events or live position) when predicting final results can leak the answer. Rigorous studies avoid this by using only truly prior information for predictions. Similarly, when predicting odds movements, one must avoid using future time steps' data in the features. Data leakage can also occur if one uses horse identity in a naive way (e.g., if a horse appears multiple times in a dataset split, a model might "remember" its past outcomes). Proper cross-validation that groups by race or time is necessary to prevent overly optimistic results.
- **Overfitting to Historical Data:** Horse racing data can be noisy and high-dimensional (many potential input factors, relatively few races for each combination of conditions). It's easy for a complex model to pick up spurious patterns that don't generalize. The systematic review of ML in betting notes that overfitting is a "*common challenge*" and is especially likely in sports with limited data or non-stationary processes ³⁸ ³⁹. Researchers mitigate this with techniques like cross-validation, regularization, and keeping models relatively simple. For example, feature selection was used by Gupta & Singh (2024) to whittle down 56 attributes to a meaningful subset, which improved generalization ⁴⁰. If one doesn't guard against overfitting, a model might show great backtest profit but then fail miserably going forward. This issue is akin to the "curve fitting" problems seen in quantitative finance.
- **Ignoring Transaction Costs and Constraints:** Many academic papers simplify the betting scenario, sometimes **ignoring Betfair commission** (which is typically 2-5% on winnings) or assuming unlimited liquidity at the posted odds. These assumptions can turn a seemingly profitable strategy into a losing one in reality. For instance, Dzalbs et al.'s strategy had a huge 83% ROI on paper ²¹, but we should ask: did they account for commission? If not, the real ROI would be somewhat lower (though still high). Commission particularly hurts high turnover strategies like trading (where you win many small margins). **Liquidity** is another cost – as Santos found, you might identify an arbitrage or value bet, but **if your stake is large, the act of placing the bet can move the odds** or only partially match, eroding the edge ⁴¹. Advanced simulations or backtests attempt to model this (e.g., only bet up to a certain amount available at the target odds, and consider the next price for remainders). Many published works, however, assume you can bet a fixed small stake at the exact odds recorded in the historical data – an ideal that might not hold if you try to bet big or if the market was thin.

- **Bet Timing and Latency:** In trading strategies, a few seconds can make the difference between a great fill and a missed opportunity. Models that rely on very fast reactions to odds changes need to operate on low-latency data and execution. If one naively uses historical data (which might be timestamped to the second or larger intervals) to simulate a strategy, they might not realize that in reality there is a **delay in Betfair bet placement (usually ~1 second in-play and near instant pre-play)** and data retrieval. By the time your model sees an odds change and signals a bet, the market may have moved. Thus, strategies that seem to capitalize on quick swings could fail when deployed. One paper noted using a 50-millisecond data feed from Betfair for analysis ⁴², acknowledging that very fine-resolution data is needed to study microstructure – but few have access to that for live use due to API limits.
- **Chasing the Market (Reflexivity):** A common failure mode is essentially chasing public money. For example, one might think “steamers (horses with shortening odds) are good bets because the odds movement indicates insider confidence.” However, by the time the odds have moved significantly, the horse is now *expensive* to bet (low odds). The earlier value is gone. The Betfair data science tutorial clearly demonstrated that reacting to moves after they happen yields no profit ³⁰. Models must anticipate moves *before* they become obvious or find horses whose odds are wrong **given all available info** – not just follow price momentum blindly. If a model simply learns “bet horses whose odds dropped from morning to noon,” it will not beat the closing price in the long run (since BSP embodies that drop already). This is a pitfall of many naive approaches, and it ties into market efficiency: only *unexpected* information can yield an edge, not known trends.
- **Unrealistic Betting Assumptions:** Some studies assume fractional betting or always betting at the best price with no risk. In practice, if you put a large bet at the best back price, you might eat through multiple price levels (getting a worse average price). If you bet at BSP, you might not know your exact odds until the off (though BSP is fair in that it’s an equilibrium price). Others assume **Kelly staking** without acknowledging variance and bankroll requirements – Kelly can maximize growth in theory but also leads to wild swings and ruin if the edge is overestimated. If a model’s edge is even slightly less than assumed, Kelly betting can bankrupt you. Thus, strategies need to be more conservative than an ideal simulation might suggest. Many punters clubs opt for flat stakes or proportional betting to manage risk, even if that’s theoretically suboptimal, because it’s more robust to model error.
- **Data Quality and Coverage:** Horse racing data can be messy – non-runners, race condition changes, etc. Betfair data itself can have gaps (e.g., API outages or missing records). Models trained on ideal datasets may encounter issues when faced with messy real-time data. There’s also the issue of **survivorship bias**: using only races that went off, excluding cancelled races or horses that were scratched after markets formed – one must be careful in how training data is compiled.
- **Human and Behavioral Factors:** While not a flaw in the model per se, when models are used by humans, **psychological biases** can become a pitfall. For example, a punters club might abandon a good model after a short losing streak (common in value betting on outsiders) because it “feels wrong,” thus never realizing the long-term edge. Academic models don’t consider this, but real-world use must. Additionally, in deployment, one must be wary of the model influencing the market: if a strategy is widely adopted or if one syndicate bets large amounts on model selections, those bets themselves will shift the odds, potentially nullifying the edge. This is analogous to high-frequency trading where an edge can disappear once too many exploit it.

Overall, the lesson from literature is to **build a margin of safety** into any betting model. Account for costs, test on truly out-of-sample data, and be skeptical of high returns without explanation. Many a promising algorithm has floundered due to one of the above pitfalls. The best studies explicitly discuss these issues – for instance, mentioning that a strategy failed when scaled up due to liquidity limits ⁴¹, or demonstrating that their results hold after commission and including realistic bet execution assumptions ⁴³ ⁴⁴. By learning from past mistakes, new research and implementations can avoid repeating them.

Adapting Models for Practical Use

Bridging the gap from research to real-world betting (whether personal automated trading or group betting) requires additional considerations. Several strategies have emerged to make ML models *usable in practice*:

- **End-to-End Automation:** For individual traders and advanced bettors, integrating the model with live data feeds and execution APIs is key. Recent work has started to provide full pipelines. Gonçalves et al. (2025) not only proposed new deep learning architectures but also *“presents a complete end-to-end framework for automated feature engineering and market interactions using the developed models in production.”* ³⁷. This means the model can live-update features (like computing indicators from the live Betfair market), make predictions, and automatically place bets or trade out positions. Such systems must be optimized for speed (so the model might be simplified enough to run in real-time) and reliability (handling API errors, etc.). Automation allows capitalizing on small edges at scale and removes human emotional interference. However, it also needs failsafes – e.g., cutting losses if the model starts misbehaving, and not risking more than a set limit.
- **Real-Time Decision Making:** Models intended for live betting (especially in-play trading) often use **rolling predictions**. For example, an LSTM model might update the predicted probability of a horse winning every few seconds during the race, adjusting bets accordingly. In practice, there’s a trade-off between model complexity and timeliness. Simpler models (or pre-computed lookup tables) might be used when ultrafast decisions are needed. Øvregård’s tennis trading model used multiple ANNs to evaluate state during matches and even created a custom cost function to be more responsive ⁴⁵ ²⁷. They achieved a tiny positive yield, showing that **responsive models coupled with fast execution can grind out profit**, but it’s technically demanding. For horse racing, in-play modeling is very tough due to race complexity and fast race conclusions (most races last only 1-3 minutes). Thus, practical focus has been more on **pre-race** trading (e.g., last 10 minutes before the off) and **immediate post-off hedging** (like taking a position before the race and trading out at BSP or soon after). These scenarios still require quick data processing but are more feasible than mid-race for horse racing.
- **Integration with Human Workflows:** For settings like punters clubs, the model output needs to mesh with how the group operates. If a club meets once a week to decide weekend bets, the model might be run beforehand to produce a report – effectively acting as a data-driven tip sheet. The presentation could be a simple table: each race with the model’s pick and an advised stake, along with key reasons (e.g., value % or top contributing factors). If the club bets through traditional bookmakers or the TAB, the model must be aware of differences (e.g., if using Best Tote or promotions, maybe they sometimes get better than BSP on certain bets – the model’s definition of “beating BSP” might need adjustment). Additionally, **group decision support** might involve giving multiple options: e.g., “In Race 5, the model identifies two value bets: #3 at 6/1 and #8 at 20/1. It

suggests a smaller bet on #8 for higher payout.” This allows the group to discuss and feel involved, using the model as a guide rather than an oracle.

- **Risk Management and Betting Strategy:** Implementations often bake in staking strategies. While an academic paper might test flat stakes for simplicity, real bettors will consider varying bet size. Some models output a suggested stake via formulas like Kelly or proportional betting based on the edge. For example, Lessmann’s approach explicitly used a Kelly strategy to capitalize on the Random Forest probabilities ⁴⁶. In practice, pure Kelly is aggressive, so bettors use fractional Kelly or limit bet sizes to manage volatility. Automated systems might have a bankroll management module ensuring no single race can bankrupt the system (stop-loss rules, max exposure limits per race, etc.). For punters clubs, which often have a fixed weekly outlay, the model can help allocate that budget across bets optimally (e.g., more on higher edge bets). **Simulation and backtesting** are employed to set these parameters: before deployment, one would simulate years of betting with the model’s historical predictions to see drawdowns and risk metrics.
- **Adaptive Learning:** The betting landscape isn’t static – models can degrade as conditions or market behavior changes. A practical system might include periodic retraining or self-correction. Some research has pointed to treating betting like a **portfolio that needs rebalancing** ⁴⁷. Future directions suggest adaptive models that learn from recent performance and maybe even weigh bets like assets (to maximize Sharpe ratio, for instance, balancing risk vs return). While this is cutting-edge, a simpler form is already used by some: e.g., dynamically adjusting a model’s output probabilities to calibrate to recent market biases, or ensemble methods that can shift weight from one sub-model to another if one starts performing better. For a punters club, adaptation might be manual – they might review model picks each month and adjust the model or rules if certain patterns of error are noticed (for example, “the model is bad at wet tracks – let’s tweak that”).
- **Use of Synthetic Data and Simulations:** Because acquiring extensive training data from Betfair (especially with detailed market depth and every tick) can be costly, researchers have used simulators like the **Bristol Betting Exchange (BBE)** to generate data ¹⁷. While simulated data isn’t a perfect replica of real markets, it can be useful for *initial training* or strategy exploration. An ML model might be pre-trained on millions of simulated betting situations to learn general patterns, then fine-tuned on real data. Caution is needed as a strategy that works in simulation might exploit artifacts of the sim that don’t hold in reality, but it’s an interesting avenue to get around data sparsity. In practice, large betting syndicates sometimes **paper trade** (simulate betting) new models in live markets to see how they would perform without risking money, before deploying fully. This is a form of using simulated execution on real data – e.g., “shadow” the bets and record outcomes to verify the edge holds when accounting for live conditions.
- **Regulatory and Logistical Constraints:** A practical consideration for social betting is the legal and platform aspect. If a club uses TAB or bookies, the model’s recommended bets might need adjustment because you can’t always get on at the desired odds (bookies might have lower odds than Betfair for longshots, for example). The model might need to scrape or accept odds from multiple sources and choose the best (odd-shopping automation). Additionally, bookmakers may limit accounts that win too much. A club might distribute bets across members’ accounts or switch to the exchange if they become too successful – these are human decisions but guided by the model’s performance. From a design perspective, a model deployed in a tool (say a web dashboard for the

club) might allow toggling different scenarios: with commission, without, with certain odds boosts, etc., so the users can see the sensitivity of profit.

In summary, making a horse racing ML model *work in practice* means **going beyond the algorithm** – one must engineer the whole system around it. This includes data pipelines for real-time odds, betting execution logic, stake sizing, result tracking, and user communication. The most successful implementations in the literature are those that didn't stop at prediction but carried through to a betting simulation or live trial, uncovering the real-world challenges. For example, Gonçalves et al. not only improved model accuracy with convolutional attention, they also **deployed the models and observed actual trading performance** ¹⁴. Such holistic approaches are crucial. As a final note, it's often said that in betting *"the last 5% of effort (the execution) makes 95% of the difference."* A mediocre model executed flawlessly with sound bankroll management can beat a great model that's implemented poorly. The research indicates this truth – the devil is in the details of turning predictions into profits.

Conclusion

The application of machine learning to horse racing betting has evolved from simple models that mimic traditional handicapping to complex systems that analyze live exchange data and automate trades. **Betfair exchange data** provides a rich playground of features – from odds movements and volumes to crowd-sourced probabilities like BSP – and ML models have leveraged these to find subtle inefficiencies. Common approaches include ensemble classifiers for predicting race outcomes and deep neural networks for forecasting price moves, often combined with domain knowledge. The literature shows that while **horse racing markets are quite efficient**, there are still edges to be found: several studies document **positive ROI strategies** under realistic conditions, although often these edges are modest or situational.

Crucially, the journey from promising model to actual profit is fraught with pitfalls. Issues like data leakage, overfitting, ignored transaction costs, and market impact can turn a winning backtest into a losing venture. Successful research and implementations tend to explicitly address these, using techniques like rigorous validation, including commission in profitability metrics, and testing in live or simulated betting environments.

For **practical use**, whether by an individual trader or a punters club, considerations of **usability and explainability** come to the forefront. Ensemble methods and rule-based overlays can improve both performance and acceptance – for example, focusing on high-value bets (even at the cost of lower strike rate) has been shown to maximize ROI ³² ³³, and providing human-interpretable reasons for selections can build trust. Models have been adapted into end-to-end systems that automatically fetch data, generate betting decisions, and execute trades, as well as into decision-support tools that augment human bettors' analysis.

The literature suggests a few recommendations for those venturing into this field:

- **Use diverse data features** (market signals *and* horse fundamentals) to cover all angles of why a horse might be mispriced, but be careful to avoid using information that is already reflected in the odds unless your goal is calibration.
- **Choose ML models appropriate to the task:** tree-based and ensemble models excel at outcome prediction with tabular data, while deep learning can uncover patterns in sequential betting data

(though at a cost of complexity). Simpler models with clear inputs may generalize better and are easier to explain to stakeholders.

- **Measure success in betting terms, not just accuracy.** Key metrics are ROI, profit on turnover, and how often your model's implied probabilities beat the market's (BSP) – these indicate real edge. Always test strategies with realistic assumptions (commission, liquidity limits).
- **Incorporate strategy and risk management into the model design.** An ML model's output is just a prediction; the decision of *when and how to bet* based on it is equally important. Techniques like thresholding for value, Kelly-fraction staking, or ensembles voting can significantly impact profitability ²² ²³ .
- **Be mindful of pitfalls:** if something looks too good (90% winners or huge ROI), double-check for leakage or biased sampling. Run backtests over extensive data and time periods. Analyze losing periods to understand if the model failed or just variance.
- **Continuously learn and adapt.** Markets may change – e.g., if many people deploy similar ML models, the inefficiencies they exploited may disappear. Having a pipeline to retrain on new data, and monitoring performance (perhaps even A/B testing model updates), is important for long-term success.

In conclusion, machine learning can indeed aid horse racing betting, but it is not a get-rich-quick button. The best results come from blending data-driven insights with betting savvy: knowing the limitations, accounting for the real-world factors, and ensuring the model's suggestions are executed in a disciplined manner. The literature so far provides a solid foundation of what works and what doesn't. Going forward, we expect to see more **integrated approaches** – for instance, reinforcement learning that directly optimizes betting profit, or portfolio approaches treating bets like investments – as hinted by recent reviews ⁴⁷ . But even as techniques advance, the core challenge remains the same: find something the market **doesn't fully know**, and bet on it wisely. Machine learning is one tool to do so, and when applied rigorously, it has proven capable of squeezing out that elusive edge in horse racing markets.

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